

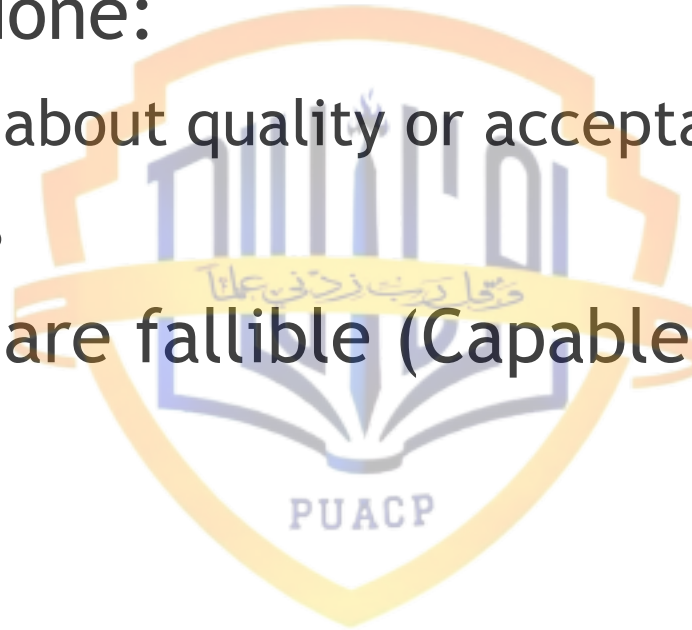
The logo of the Punjab University of Applied Sciences and Technology (PUACP) is centered in the background. It features a shield-like emblem with a yellow border. Inside the shield, there is a stylized open book with a flame above it, and the acronym 'PUACP' at the bottom. A yellow banner with Urdu text is draped across the middle of the shield.

Software Quality Assurance

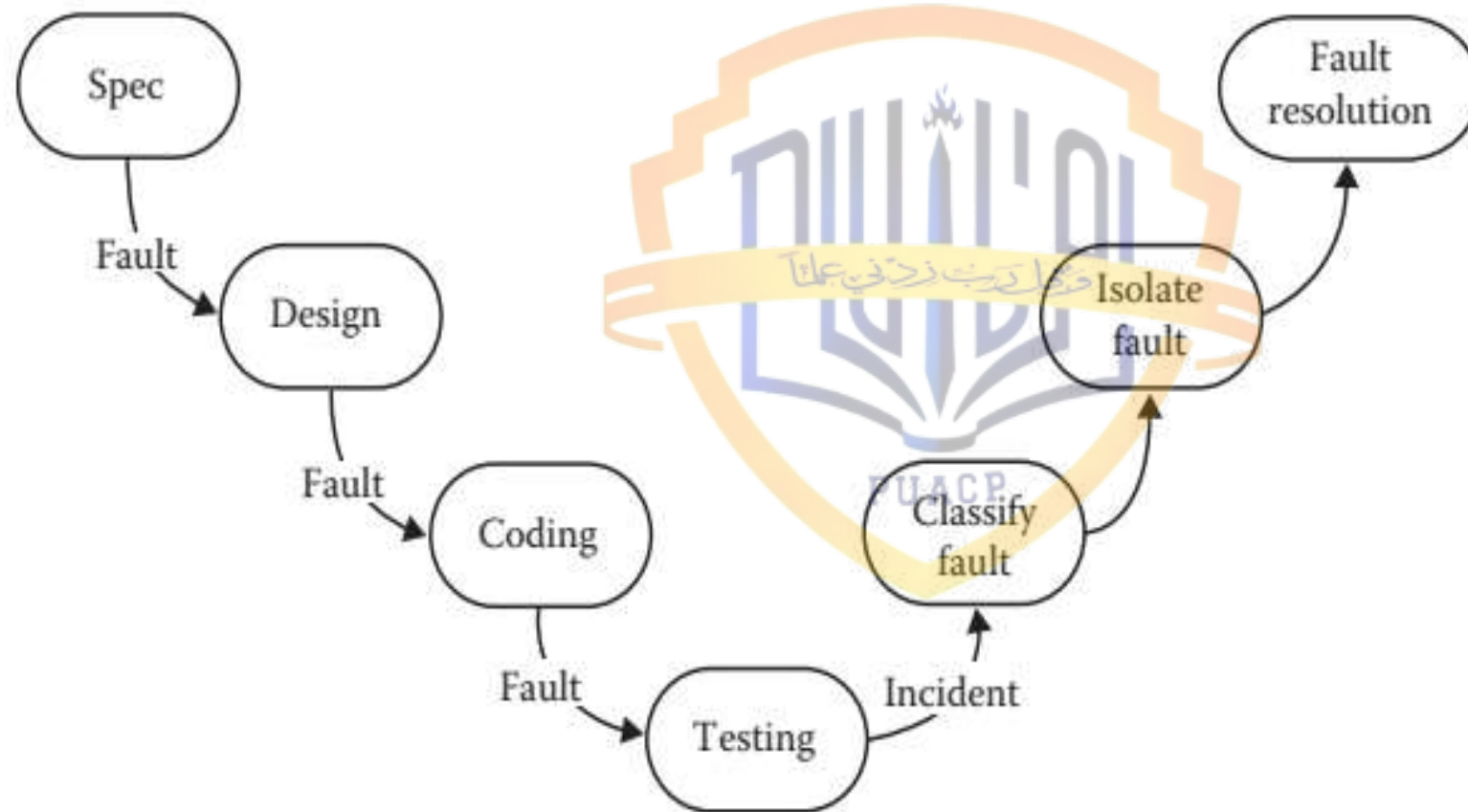
By: Muhammad Fahim

A Perspective on Testing

- Software Testing is done:
 - ❑ To make a judgment about quality or acceptability
 - ❑ To discover problems
- We test because we are fallible (Capable of making mistakes)



A Testing Life Cycle



Basic Definitions

➤ Error

- ❑ People make errors/mistakes
- ❑ Coding errors are called bugs
- ❑ Errors tend to propagate (requirements to design and coding)
- ❑ Errors of omission vs errors of commission

➤ Fault

- ❑ Result of an error/representation of an error (narrative text, UML modeling diagram, hierarchy charts and source code)
- ❑ Faults of omission vs faults of commission
- ❑ Faults of omission are more difficult to detect and resolve

Basic Definitions (Continued)

➤ Failure

- ❑ Occurs when code corresponds to a fault executes
 - Failures in executable code
 - Failures due to faults of commission
 - How to deal with faults of omission (Faults that never occur or do not occur for a long time)

➤ Incident

- ❑ Symptom associated with a failure (Failure may not be readily apparent to user i.e. customer or tester)

Basic Definitions (Continued)

➤ Test

- ❑ Concerned with errors, faults, failures and incidents
- ❑ Act of exercising software with test cases
- ❑ Goals of Tests
 - To find failures
 - To demonstrate correct execution

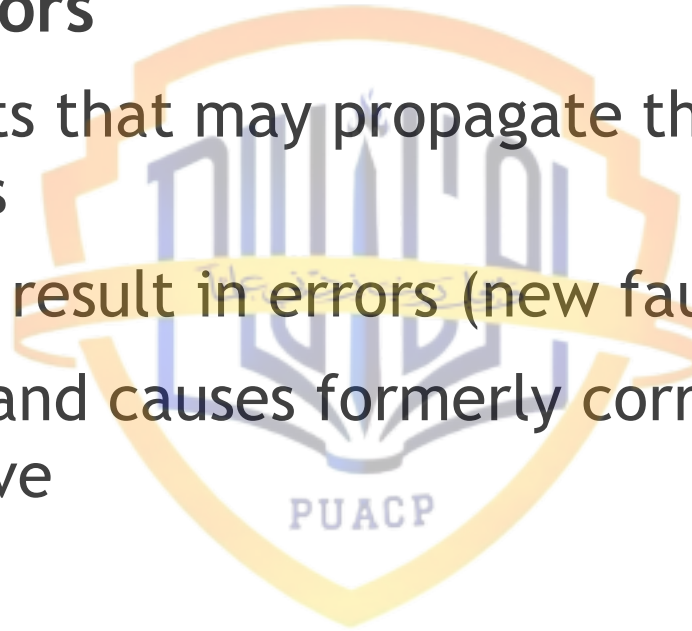
➤ Test Case

- ❑ Has an identity and associated with program behavior
- ❑ Has a set of inputs and expected outputs

Basic Definitions (Continued)

➤ Opportunities for Errors

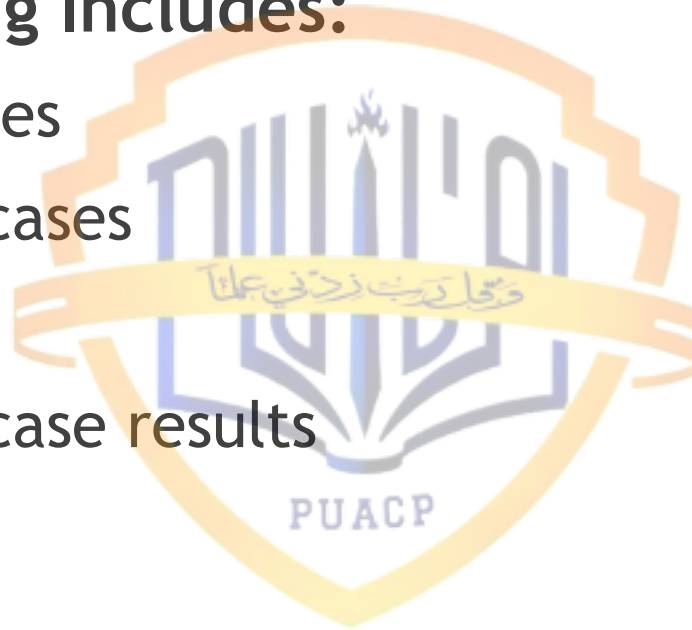
- ❑ Errors results in faults that may propagate through development process
- ❑ Fault resolution may result in errors (new faults)
- ❑ The fix is deficient, and causes formerly correct software to misbehave



Test Cases (The Core of Testing)

➤ The Process of Testing Includes:

- ❑ Planning the test cases
- ❑ Developing the test cases
- ❑ Running test cases
- ❑ Evaluating the test case results



Test Cases (A Recognized Work Product)

➤ A Test Case Must have:

- ❑ Test case identity
- ❑ Brief statement of purpose
- ❑ Running test cases (e.g. a business rules)
- ❑ Description of pre-conditions
- ❑ Actual test case inputs
- ❑ Expected outputs
- ❑ Description of expected post conditions
- ❑ Execution history (date of test run, person who ran it, version on which it was run, the pass/fail result)
- ❑ Output is frequently overlooked



Venn Diagram (An Insight)

➤ A Test Case Must have:

- ❑ Testing is fundamentally concerned with behavior
- ❑ Behavior is orthogonal to code-based view
- ❑ Code-based view focuses on what it is?
- ❑ Behavioral-view is concerned with what it does?
- ❑ Based documents are written by and for developers; emphasis is code-based instead of behavior-based

Venn Diagram (An Insight)

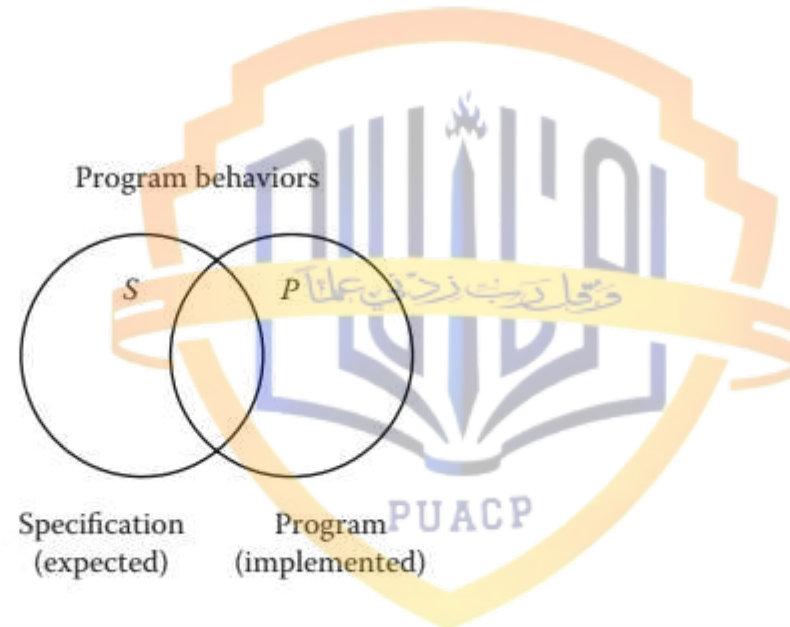


Figure 1.2 Specified and implemented program behaviors.

Venn Diagram (An Insight)

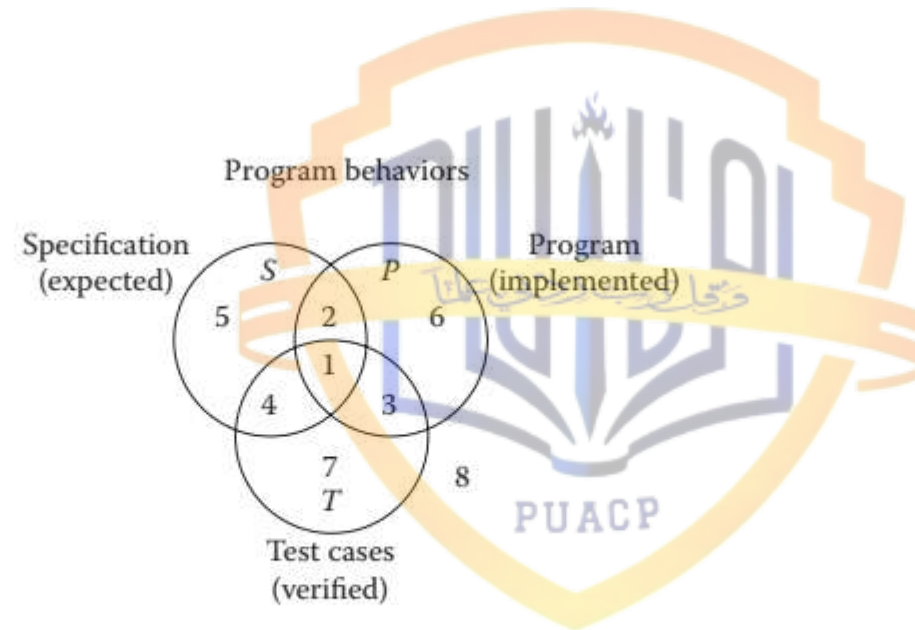


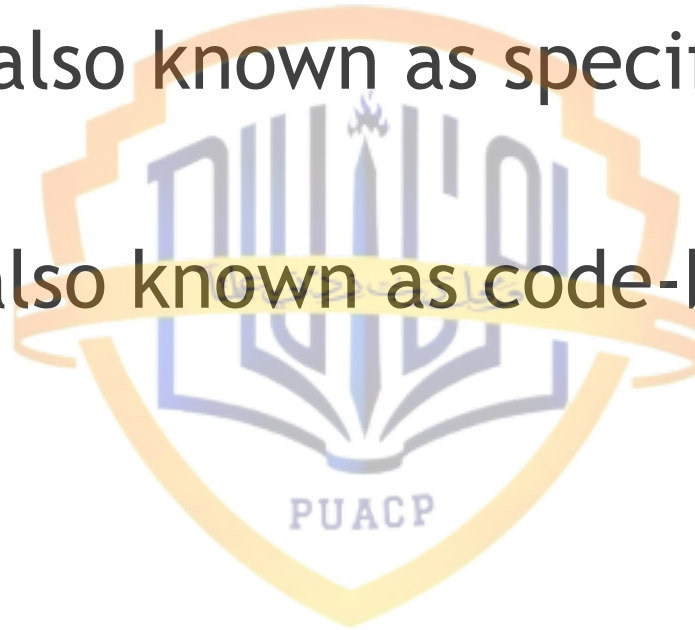
Figure 1.3 Specified, implemented, and tested behaviors.

Software Testing...

- Testing is the determination of the extent of program behavior that is both specified and implemented.
- Correctness only has meaning with respect to a specification and an implementation

Identifying Test Cases: Testing Methodologies

- Functional testing also known as specification-based testing
- Structural testing also known as code-based testing



Specification-Based Testing (Black Box Testing)

- To identify a test case only the information available in specification is used

Advantages:

- ❑ Test cases are independent of software implementation
- ❑ Test case development and implementation can occur parallelly

Disadvantages:

- ❑ Redundancies may occur between test cases
- ❑ May have the gaps of untested software